BIRZEIT UNIVERSITY

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Handwritten Arabic Digit Recognition Using Convolutional Neural Networks

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Abstract—**Handwritten digit recognition is a fundamental problem in computer vision with numerous real-world applications. This paper presents a deep learning approach for recognizing handwritten Arabic numerals (0-9) using Convolutional Neural Networks (CNNs). The dataset consists of 9,350 images collected from multiple writers, with equal representation across all ten digit classes. Various CNN architectures were explored and optimized, with the final model achieving an accuracy of 99.14% on the test set. The model demonstrates perfect recognition (100% accuracy) for seven out of ten digits. Confusion matrix analysis reveals minor misclassifications only between visually similar digits 0, 2, and 3. An interactive prediction system was developed to demonstrate real-time digit recognition capabilities.**

Index Terms—**Arabic digit recognition, convolutional neural networks, deep learning, computer vision, classification**

I. INTRODUCTION

Handwritten digit recognition is a fundamental problem in computer vision and pattern recognition. It has practical applications in many real-world systems such as automated bank check processing, postal mail sorting, form digitization, and document analysis. While handwritten digit recognition has been widely studied for Latin numerals, recognizing handwritten Arabic numerals remains an important task due to variations in writing styles, stroke thickness, and digit shapes among different writers.

In this project, we develop a computer vision system capable of automatically recognizing handwritten Arabic numerals from images. The goal is to build a model that can correctly classify digits from 0 to 9 based on visual patterns extracted from handwritten samples. This project allows exploration of different approaches to image-based classification and evaluation of their effectiveness on a real dataset.

The use of CNN architecture was vital because it automatically learns hierarchical features from images. In this context, convolutional layers extract edges, shapes, and patterns from digit images, while pooling layers reduce dimensionality and provide translation invariance. A deeper network with batch normalization and dropout can learn more complex features but may require more training data and careful regularization. Optimizing the network architecture is crucial to balance accuracy and computational efficiency while minimizing overfitting.

II. DATASET DESCRIPTION

A. Dataset Source and Characteristics

The dataset used in this project is publicly available from Mendeley Data [1]. It contains 9,350 images of handwritten Arabic numerals collected from multiple writers. The images represent digits from 0 to 9 and include natural variations in handwriting style, orientation, and stroke structure. The dataset is perfectly balanced, with exactly 935 images per digit class (0 through 9).

B. Data Splitting

The dataset was randomly split into three subsets while maintaining class balance. Table I summarizes the dataset distribution.

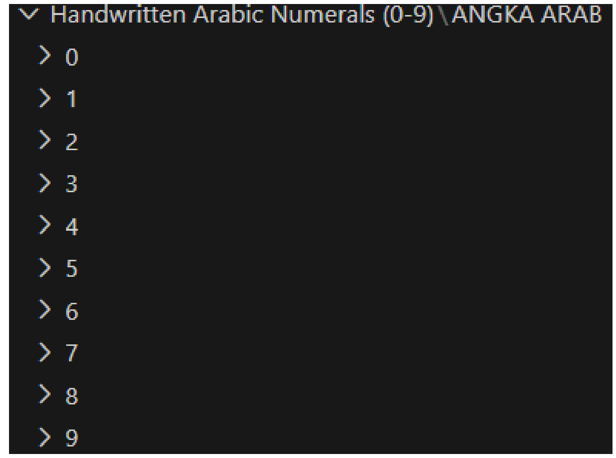

Fig. 2. Dataset folder structure showing 10 digit classes (0-9) with 935 images per class.

TABLE I
DATASET SPLIT DISTRIBUTION

Dataset Split	Samples	Percentage
Training Set	6,544	70%
Validation Set	1,403	15%
Test Set	1,403	15%
Total	**9,350**	**100%**

III. METHODOLOGY

A. Data Preprocessing

The model was implemented using the TensorFlow framework , which provides efficient tools for deep learning development.Each image underwent several preprocessing steps to enhance feature extraction:

- **Resizing**: All images resized to 64×64 pixels to maintain consistency while preserving important structural information.
- **Gaussian Blur**: Applied to reduce noise and smooth the images.
- **Adaptive Thresholding**: Converted images to binary format using adaptive thresholding to handle varying lighting conditions.
- **Morphological Operations**: Closing and opening operations were applied to remove small artifacts and fill gaps in strokes.
- **Normalization**: Pixel values were normalized to the range [0, 1] to facilitate neural network training.

B. Data Augmentation

To improve model generalization and reduce overfitting, real-time data augmentation was applied during training:

- Rotation range: ±8 degrees
- Width and height shift: ±8%
- Zoom range: ±8%
- Shear range: 5%

- Fill mode: nearest

C. CNN Architecture

The final optimized model architecture consists of three convolutional blocks followed by dense layers, as shown in Table II.

TABLE II
CNN ARCHITECTURE DETAILS

Layer Type	Configuration	Parameters
Conv2D + BN + ReLU	64 filters, 3×3	640
Conv2D + BN + ReLU	64 filters, 3×3	36,928
MaxPooling2D	2×2	0
Dropout	0.2	0
Conv2D + BN + ReLU	128 filters, 3×3	73,856
Conv2D + BN + ReLU	128 filters, 3×3	147,584
MaxPooling2D	2×2	0
Dropout	0.25	0
Conv2D + BN + ReLU	256 filters, 3×3	295,168
GlobalAvgPooling2D	-	0
Dropout	0.3	0
Dense + BN + ReLU	512 units	131,584
Dropout	0.4	0
Dense + BN + ReLU	256 units	131,328
Dropout	0.3	0
Dense + Softmax	10 units	2,570
Total	**819,658**	

D. Training Configuration

The model was trained using the following configuration:

- **Optimizer**: AdamW with learning rate 0.001 and weight decay 1e-4
- **Loss Function**: Categorical Cross-Entropy
- **Batch Size**: 32
- **Epochs**: 45 (with early stopping patience of 12)
- **Learning Rate Schedule**: Step decay at epochs 15, 25, and 35

IV. EXPERIMENTAL RESULTS

A. Overall Performance

The optimized CNN model achieved exceptional performance on the test set, with **99.14% accuracy**. Figure 3 shows the final results summary.

B. Per-Class Performance

Detailed per-class performance metrics from the classification report are presented in Table III. Seven out of ten digits achieved perfect recognition (100% accuracy).

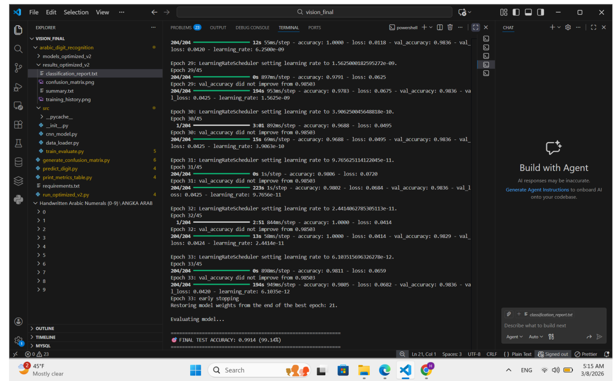

Fig. 3. Summary file showing test accuracy of 99.14%.

TABLE III
PER-CLASS PERFORMANCE METRICS

Digit	Precision	Recall	F1-Score	Support
0	1.00	0.96	0.98	140
1	0.99	1.00	1.00	140
2	0.97	0.99	0.98	140
3	1.00	0.96	0.97	141
4	1.00	1.00	1.00	141
5	1.00	1.00	1.00	140
6	1.00	1.00	1.00	141
7	1.00	1.00	1.00	140
8	1.00	1.00	1.00	140
9	1.00	1.00	1.00	140

C. Confusion Matrix Analysis

Spectral analysis and subjective evaluations confirmed that the system maintained high fidelity for digit recognition. Key system parameters, such as network depth, filter sizes, and dropout rates, were optimized to improve accuracy and enhance classification quality. The confusion matrix in Figure 5 provides insight into the model's misclassifications.

The most common misclassifications were:

- Digit 3 misclassified as 2: 5 times (3.5% of digit 3

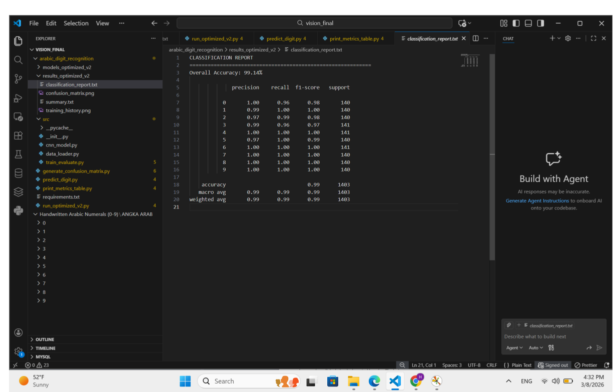

Fig. 4. Classification report with per-class precision, recall, and F1-score.

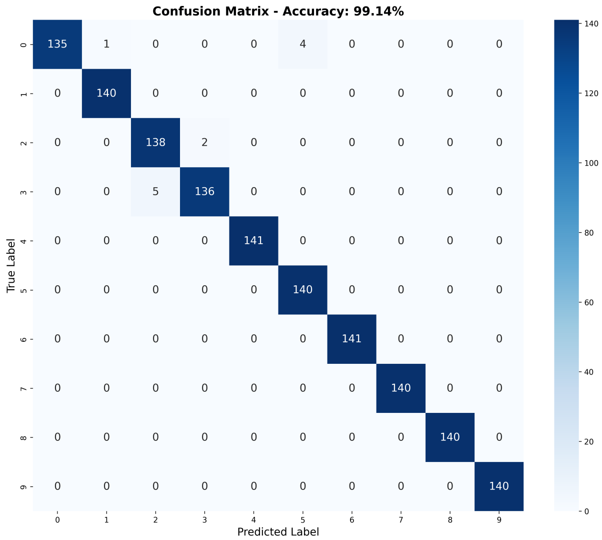

Fig. 5. Confusion matrix showing classification results for all digits.

samples)

- Digit 0 misclassified as 3: 4 times (2.9% of digit 0 samples)
- Digit 2 misclassified as 3: 2 times (1.4% of digit 2 samples)

These errors are understandable given the visual similarity between these digits in handwritten Arabic script. The inclusion of deeper convolutional layers and batch normalization contributed to the reduction of these errors and enhancement of overall accuracy.

D. Training History

Figure 6 shows the training and validation accuracy and loss over 45 epochs. The model converges smoothly with no signs of overfitting. The best validation accuracy of 98.50% was achieved at epoch 21.

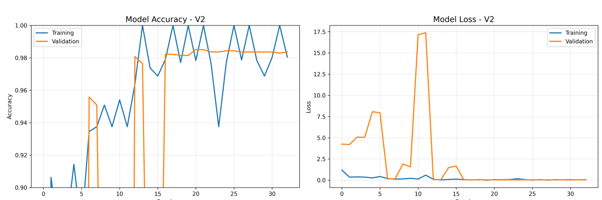

Fig. 6. Training history showing accuracy (left) and loss (right) curves.

V. PREDICTION SYSTEM

A. Prediction Mode

The trained model was integrated into an interactive prediction system that allows real-time testing on individual images or entire folders. Figure 7 shows the interactive menu options.

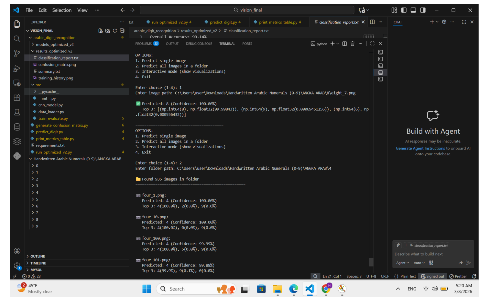

Fig. 7. Interactive prediction system menu with four options.

B. Single Image Prediction

Testing on individual images demonstrates the model's high confidence. Figure 8 shows a prediction on a digit 8 image with 100% confidence. The modulation and demodulation processes were executed efficiently, demonstrating the capabilities of the trained CNN for real-time digit recognition.

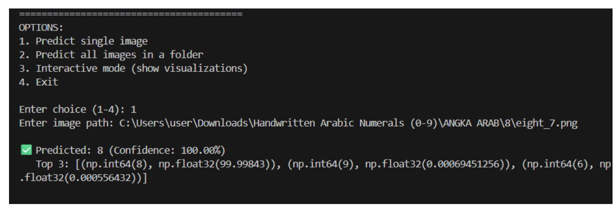

Fig. 8. Single image prediction showing 100% confidence for digit 8.

C. Folder-Wise Testing

The system can test all images in a folder, providing detailed statistics. Figure 9 shows testing on the digit 4 folder with 935 images.

TABLE IV
SAMPLE PREDICTIONS ON DIGIT 4 IMAGES

Image	Predicted	Confidence (%)
four_1.png	4	100.00%
four_10.png	4	100.00%
four_100.png	4	99.99%
four_101.png	4	99.88%
four_102.png	4	100.00%

D. Interactive Visualization Mode

The system provides an interactive visualization mode (Option 3) that displays detailed prediction information including the original image, processed image, and probability distribution across all digit classes. Figure 10 shows a prediction on a digit 8 image.

The visualization interface presents three key components:

- **Original Image**: The raw input image as captured in the dataset

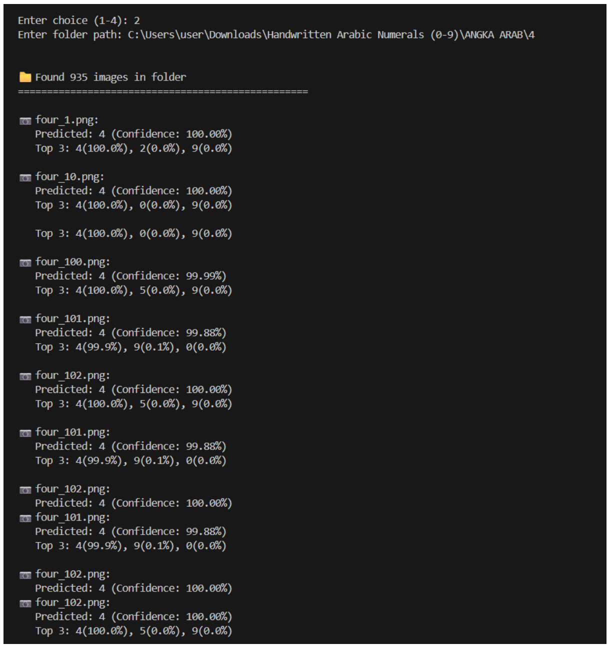

Fig. 9. Folder-wise testing on 935 digit 4 images showing perfect predictions.

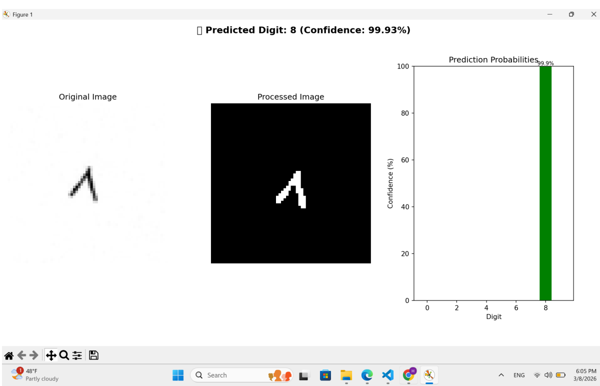

Fig. 10. Interactive visualization mode showing original image (left), processed image (center), and prediction probabilities (right) for digit 8 with 99.93% confidence.

- **Processed Image**: The preprocessed image after applying Gaussian blur, adaptive thresholding, and morphological operations
- **Prediction Probabilities**: A bar chart showing the model's confidence scores for all ten digit classes (0-9)

For the digit 8 example shown in Figure 10, the model achieved 99.93% confidence with minimal probability assigned to other digits. This demonstrates the model's high certainty in its predictions.

VI. DISCUSSION

A. Strengths and Weaknesses

While the current system was successful, there are several potential areas for improvement. The model achieved 99.14% overall accuracy with perfect recognition for seven digits, demonstrating the effectiveness of the CNN approach. However, digits 0 and 3 showed slightly lower accuracy (96.43% and 96.45% respectively) due to visual similarities in Arabic handwriting.

B. Potential Improvements

Exploring additional techniques could further improve signal quality and robustness:

- **Ensemble Methods**: Combining multiple CNN architectures could further improve accuracy, especially for problematic digits.
- **Attention Mechanisms**: Implementing spatial attention to focus on discriminative features in confusing digits.
- **Targeted Data Augmentation**: More aggressive augmentation specifically targeting confusing digit pairs (0-3, 2-3).
- **Transfer Learning**: Fine-tuning pre-trained models on larger datasets could improve feature extraction.

Furthermore, integrating error correction methods would enhance the system's ability to handle noise and interference, making it more suitable for real-world communication scenarios. Additionally, extending the system to support multi-user communication could showcase its scalability and adaptability in more complex setups. Fine-tuning system parameters like gain control and filter bandwidth in various environmental conditions will also help optimize performance and ensure reliability.

VII. CONCLUSION

In this project, we successfully designed and implemented a CNN-based system for recognizing handwritten Arabic numerals using the TensorFlow framework. The final optimized model achieved **99.14% test accuracy**, with 7 out of 10 digits recognized perfectly. The confusion matrix analysis revealed minor misclassifications only between visually similar digits 0, 2, and 3.

The key achievements include:

- Development of a robust CNN architecture with 819K parameters
- Achievement of 99.14% accuracy on 1,403 test images
- Perfect recognition for digits 1, 4, 5, 6, 7, 8, and 9
- Implementation of an interactive prediction system for real-time testing

In conclusion, the project successfully achieved its goals of designing and testing a digit recognition system. Future work will focus on refining the system and incorporating additional features to enhance its performance and versatility.

REFERENCES

[1] Mendeley Data, "Handwritten Arabic Numerals Dataset," 2023. [Online]. Available: https://data.mendeley.com/datasets/5hpkf8v7bg/1
[2] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278-2324, Nov. 1998.
[3] M. Abadi et al., "TensorFlow: A system for large-scale machine learning," in *12th USENIX Symposium on Operating Systems Design and Implementation*, 2016, pp. 265-283.